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Nordic strength and history of hamstring injury in Australian Football League players

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ABSTRACT

Objectives: Compare hamstring strength between Australian Football League (AFL) players with and without a prior hamstring injury and determine the effect of the number of previous hamstring injuries, time since the last injury, and injury severity, on hamstring strength.
Design: Cross-sectional, retrospective.
Setting: AFL clubs.
Participants: 124 AFL players.
Main outcome measures: Bilateral hamstring strength was assessed on a Nordbord (Vald Performance) during the Nordic Hamstring Exercise. Self-reported questionnaires were used to record previous hamstring injuries. Players were categorized into No Injury or Hamstring Injury groups. Previously injured players were subgrouped based on number of prior hamstring injuries (single or multiple), time since the last hamstring injury (≤1 or > 1-year ago), and hamstring injury severity (≤3 or > 3 matches missed).
Results: 19 hamstring injuries were reported. Hamstring strength was not different between players with and without a history of hamstring injury when assessed in absolute (N) or relative (i.e., $N.kg^{-1}$) terms. No differences in strength were detected between hamstring injury subgroups when assessed in absolute or relative terms.
Conclusions: AFL players that experienced a previous hamstring injury did not exhibit deficits in hamstring strength relative to their uninjured limb or players without a previous hamstring injury.
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1. Introduction

The incidence of hamstring injuries within the Australian Football League (AFL) (Orchard et al., 2013) and other football codes such as European football (soccer) (Ekstrand et al., 2013, 2016) and Gaelic football (Roe, Murphy, et al., 2018) remains high. It is well established that a history of hamstring injury is the biggest risk factor for a future hamstring injury in AFL players (Orchard et al., 2020) and other athletic populations (Green et al., 2020). Furthermore, this elevated risk is greatest when the previous hamstring injury is considered recent (i.e., either ≤8 weeks (Orchard et al., 2020), or within the same playing season (Green et al., 2020)). Hamstring weakness has also been identified as a potential risk factor (Opar, Williams, et al., 2015) and consequence (Maniar et al., 2016) of hamstring injuries in AFL players. For example, Opar, Williams, et al. (2015) reported that AFL players with lower pre-season measures of eccentric knee flexor strength had a significantly increased risk of sustaining a future hamstring injury within the 2013 AFL season, and this risk was amplified in players with a history of hamstring injury, whereas residual deficits in strength have been reported after hamstring injury (Maniar et al., 2016). Training and rehabilitation practices within the AFL continue to evolve and it is unclear if strength deficits remain, and whether there is any association between hamstring strength and hamstring injury.

Based on a meta-analysis of three studies in athletic populations that investigated the effect of prior hamstring injury on strength during the Nordic Hamstring Exercise (NHE), henceforth referred to

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as Nordic strength, it was concluded that Nordic strength was significantly lower in previously injured limbs compared with the uninjured contralateral limb (Maniar et al., 2016). Similar findings have been reported in subsequent studies in amateur (Vicens-Bordas et al., 2020) and professional male soccer players (Ribeiro-Alvares et al., 2021), although no such asymmetry was observed in elite track and field athletes (Giakoumis et al., 2020), or in elite Gaelic footballers (Roe, Malone, et al., 2018). Given the inconsistency of these findings, it is unclear if Nordic strength deficits persist following hamstring injury and the impact of previous hamstring injury on Nordic strength cannot be confirmed.

Prior studies that have investigated Nordic strength in injured and uninjured players have tended to investigate single, unilateral hamstring injuries that have occurred within the previous 12 (Opar, Willams, et al., 2015; Ribeiro-Alvares et al., 2021; Roe, Malone, et al., 2018; Vicens-Bordas et al., 2020) or 18-months (Timmins et al., 2015). However, the effect of hamstring injury on Nordic strength might be expected to depend on factors including the number of prior injury episodes, the recency of the injury, and the severity of the injury. This information is important for the treating clinician as a reasonable clinical assumption may be that players with multiple or more severe injuries may exhibit greater residual strength loss, and therefore require longer and/or more specialized rehabilitation. To our knowledge, no studies have collectively investigated the impact of different variables of hamstring injury, i.e., number of episodes, the recency of the injury, and the severity of the injury on Nordic strength in AFL players. The close relationship of hamstring injury with subsequent injury and the possible interaction with strength is of particular interest to the clinician and warrants further investigation.

The purpose of this study was to: (1) compare Nordic strength and strength asymmetry between AFL players with and without a prior hamstring injury, and (2) determine the effect of the number of previous hamstring injuries, time since the last injury, and injury severity, on Nordic strength and between-limb asymmetry. We hypothesised that players with a history of hamstring injury would display lower Nordic strength and greater between-limb asymmetry than uninjured players, and that these deficits would be augmented in players with more recent and more severe hamstring injuries.

2. Methods

2.1. Participants

One-hundred and twenty-four AFL players were recruited from three clubs. Due to the potential impact of an orthopaedic injury to the lumbopelvic area, or lower limbs, or a history of lower limb soft-tissue injury apart from hamstring injury, on Nordic strength, any player who reported an injury of this description was excluded from the study. The study was approved by the relevant institutional Human Research Ethics Committee (approval number: 2017/896) and all participants provided written informed consent prior to participation in the study.

2.2. Injury history questionnaire

At the time of testing, all players completed a self-reported injury history questionnaire which included information regarding hamstring injuries, i.e., leg(s) previously injured, time taken to return to play, the most recently injured leg, and time since last injury. Previous hamstring injury was defined as having sustained an acute, non-contact incident to the hamstring group from AFL training or playing, as diagnosed by the club medical staff, that resulted in missing at least one AFL match. This definition is consistent with the time-loss definition used by the AFL (Orchard et al., 2013). From the injury history questionnaire, two groups were initially identified: those with no prior hamstring injury (No Injury), and those with a history of hamstring injury (Hamstring Injury). Subsequently, the hamstring injury group were further subgrouped based on three factors: (i) the 'number' of hamstring injury episodes (single or multiple episodes), (ii) 'time since last injury' ($\leq$1-year ago or > 1-year ago), and (iii) 'severity of injury' (number of matches missed; $\leq$ 3 or > 3 matches). Subgrouping based on the number of matches missed has been used by other investigators (Vicens-Bordas et al., 2020), and similarly, reflects the average duration for hamstring injury within the AFL, i.e., 3.5 matches (season incidence/season prevalence) (Orchard et al., 2013). The subgrouping relating to the period since the last hamstring injury was based on $\leq$ 1-year ago, as used previously (Opar, Willams, et al., 2015; Ribeiro-Alvares et al., 2021; Roe, Malone, et al., 2018; Vicens-Bordas et al., 2020), or > 1-year ago.

2.3. Nordic strength

Nordic strength was assessed bilaterally during performance of the NHE as described previously (Opar et al., 2013). Players knelt on a Nordbord (Vald Performance, Brisbane, Australia) with their ankles secured in unilateral ankle hooks which were attached to independent uniaxial load cells. Participants were instructed to maintain a neutral hip and pelvic alignment and to slowly lean forward while resisting the forward lean by maximally contracting the hamstrings bilaterally. Participants initially performed a warm-up consisting of less than five submaximal repetitions of the NHE, followed by three maximal repetitions. Maximum Nordic strength was defined as the maximum force measured from each leg during the three trials and was expressed in absolute terms (Newtons; N). A trial was deemed acceptable when the force output achieved maximum, followed by a rapid decline in force which occurred when the player was no longer able to resist the effects of gravity acting against the segments proximal to the knee joint. Testing was conducted at the mid-to-end of an AFL preseason (January to March n = 98 in 2017; n = 26 in 2018). Reliability of a prototype of the Nordbord used in assessing strength has been reported previously (intraclass correlation coefficient 0.83 to 0.90, typical error 21.7–27.5 N or 5.8–8.5%) (Opar et al., 2013). Nordic strength asymmetry was computed by subtracting the value of the non-dominant leg from the value for the dominant leg (preferred kicking limb) for the No injury group, and by subtracting the value for the injured limb from the non-injured limb for the Hamstring injured group and was reported in absolute terms (N).

2.4. Statistical analysis

Data were checked for normality and parametric or non-parametric statistics were used accordingly. Independent samples t-tests were used to assess the effect of group (No Injury vs Hamstring Injury) on participant characteristics (age, mass, and height), Nordic strength (dominant limb in the No Injury group and injured limb in the Hamstring Injury group) and Nordic strength asymmetry. The potential effect of age, mass, and height on Nordic strength and Nordic strength asymmetry was assessed using Analysis of Covariance. Paired samples t-tests were used to determine between limb differences in Nordic strength within each group (No Injury and Hamstring Injury). The effect of number of episodes (single versus multiple), time since last injury ($\leq$1-year ago versus > 1-year ago), and injury severity ($\leq$3 matches missed versus >3 matches missed) on Nordic strength and Nordic strength asymmetry was assessed using the Kruskal-Wallis test. For the participants within the Hamstring Injury group, between limb

differences (injured limb versus uninjured limb) in Nordic strength were assessed for each subgroup (i.e., single, multiple, ≤ 1-year ago, > 1-year ago, ≤ 3 matches missed and >3 matches missed) using the Wilcoxon signed-rank test. As Nordic strength is, in part, dependent on bodyweight (Buchheit et al., 2016), equivalent statistical analyses were conducted with relative strength, i.e., $N.kg^{-1}$. All statistical analyses were conducted using the Statistical Package for the Social Sciences (version 26, IBM, USA) and significance was accepted for $p < 0.05$. Data are presented as mean (standard deviation; SD and 95% confidence intervals; CI) for parametric statistics and median (inter-quartile range; IQR) for non-parametric statistics. Effect sizes were computed using Cohen's *d* (parametric) or Hedge's *g* (non-parametric).

3. Results

3.1. Participant characteristics

After applying the exclusion and grouping criteria to the original 124 players, 47 players reported no prior hamstring injury (n = 43 with complete strength and injury history data) and 20 players reported a prior hamstring injury (n = 15 with complete strength and injury history data). For the Hamstring Injury group, the average number of injuries was 3 (SD, 2; range 1–8), 'time since last injury' was 86 (SD, 106; range 16–468) weeks and average 'severity' was 5 (SD, 8; range 1–40) matches. Four players reported a history of bilateral hamstring injuries and both limbs were allocated to the injured limb when compared with the No Injury group, however, these four players were not included within the asymmetry analysis due to not having an uninjured side. Players with a history of hamstring injury were significantly older than those without (mean difference: 2 (1) years; 95% CI: 1–4 years, $d = 0.77$, $p < 0.01$; Table 1). There were no significant between group difference in body mass and height (Table 1), subsequently, these measures were not used in further analyses. The average 'number' of hamstring injuries for the multiple hamstring injury subgroup was 4 (SD, 2; range 2–8). The average 'time since last injury' was 33 (SD, 15; range 16–52) weeks for the ≤ 1-year subgroup and for the > 1-year subgroup was 147 (SD, 131; range 56–468) weeks. The average 'severity' for the ≤3 matches missed subgroup was 2 (SD, 1; range 1–3) and for the >3 matches missed subgroup was 8 (SD, 11; range 4–40).

3.2. Nordic strength and Nordic strength asymmetry for injured and uninjured groups

There were no significant differences in Nordic strength between the No Injury group (dominant limb) and the Hamstring Injury group (injured limb, mean difference: 5 (28) N; 95% CI: -51 to 60 N, $d = 0.05$, $p = 0.87$; Fig. 1a) or when age was used as a covariate (mean difference: 14 (30) N; 95% CI: -45 to 73 N, $d = 0.14$, $p = 0.63$; Fig. 1a). Similarly, there were no significant differences in Nordic strength asymmetry between the No Injury group and the Hamstring Injury group (mean difference: 13 (16) N; 95% CI: -19 to 45 N, $d = 0.26$, $p = 0.42$; Fig. 1b), or when age was used as a covariate (mean difference: 2 (15) N; 95% CI: -29 to 33 N, $d = 0.05$, $p = 0.88$; Fig. 1b).

Table 1
Demographic data for the No Injury and Hamstring Injury groups. Data are mean (SD).

	No Injury	Hamstring Injury	p value
N	47	20	
Age (years)	21 (3)	23 (3)[a]	<0.01
Mass (kg)	86 (9)	87 (7)	0.74
Height (cm)	189 (7)	188 (5)	0.62

[a] Denotes significantly different from the No Injury group.

3.3. Between limb differences in Nordic strength

There were no significant differences in Nordic strength between the dominant and non-dominant limb for the No Injury group (mean difference = 7 (46) N, 95% CI: -7 to 21 N, $d = 0.06$, $p = 0.35$; Table 2), or between the previously injured limb and the uninjured contralateral limb for the Hamstring Injury group (mean difference = −20 (56) N, 95% CI: -57 to 18 N, $d = 0.25$, $p = 0.27$; Table 2).

3.4. Effect of number of episodes, time since injury, and injury severity on Nordic strength and Nordic strength asymmetry

There were no significant differences in Nordic strength between the injured limb for number of previous episodes (single versus multiple; $g = 0.99$; $p = 0.25$; Fig. 2a), time since last injury (≤1-year ago versus > 1-year ago; $g = 0.61$; $p = 0.61$; Fig. 2b), and injury severity (≤3 matches missed versus >3 matches missed; $g = 0.74$; $p = 0.11$; Fig. 2c). Similarly, there were no significant differences in Nordic strength asymmetry for number of previous episodes (single versus multiple; $g = 0.57$, $p = 0.72$; Fig. 3a), time since last injury (≤1-year ago versus > 1-year ago; $g = 1.09$, $p = 0.21$; Fig. 3b), and injury severity (≤3 matches missed versus >3 matches missed; $g = 0.09$, $p = 0.71$; Fig. 3c).

Further, there were no significant differences in Nordic strength between the injured limb and the uninjured contralateral limb for the single ($g = 0.08$, $p = 0.46$) and multiple episode subgroups ($g = 0.57$, $p = 0.50$; Table 3), the ≤ 1-year ($g = 0.55$, $p = 0.12$), and > 1-year time since last injury subgroups ($g = 0.06$, $p = 0.35$), and for the ≤3 weeks ($g = 0.56$, $p = 0.23$) and >3 weeks ($g = 0.34$, $p = 0.11$) injury severity subgroups.

Equivalent statistical analyses for relative Nordic strength did not demonstrate significant findings for any test (all p values > 0.05, Supplementary Figs. 1–3, and Supplementary Tables 1–2).

4. Discussion

In contrast to our hypotheses, no significant differences in Nordic strength or strength asymmetry were identified between players with and without a history of hamstring injury when data was expressed as absolute strength (N) and relative strength ($N.kg^{-1}$). Furthermore, no effect of number of injuries, time since last injury, or injury severity on Nordic strength was detected. Overall findings of this study, therefore, suggest that AFL players that have experienced a hamstring injury do not exhibit Nordic strength deficits or asymmetry in strength at 86 (106) weeks following hamstring injury. Nordic strength in the injured limb in the present study was approximately 40% higher compared with a prior study that reported lower Nordic strength in the injured compared with non-injured AFL players (Opar, Williams, et al., 2015), which likely reflects increased emphasis on hamstring strengthening during training and rehabilitation in the AFL in recent years. The cause of ongoing hamstring injuries in the AFL is likely explained by a range of complex interacting factors that will require further investigation.

4.1. Between group differences in Nordic strength

Consistent with a recent meta-analysis investigating hamstring injury risk factors (Green et al., 2020), players who reported a history of hamstring injury were significantly older compared with

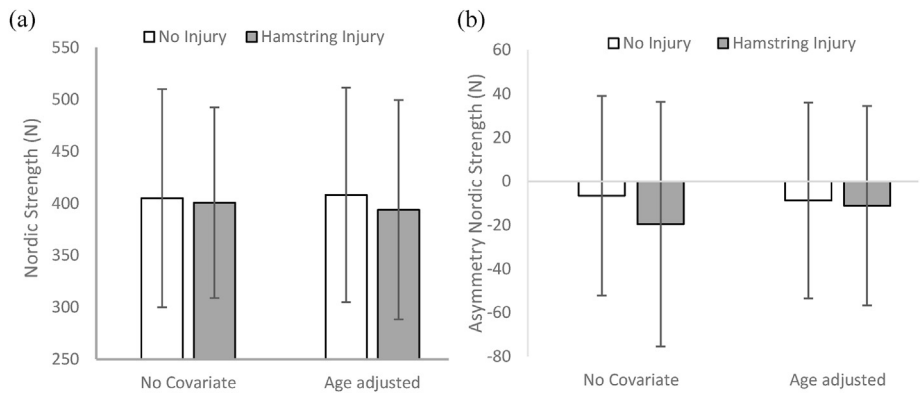

Fig. 1. Group mean and standard deviation Nordic strength for the No Injury (n = 43; dominant limb) and the Hamstring Injury (n = 19; injured limb) groups without and with age as covariate (20.79 years; a), and between limb asymmetry in Nordic strength for the No Injury (n = 43; non-dominant limb minus dominant limb) and the Hamstring Injury (n = 11; injured limb minus uninjured limb) groups without and with age as covariate (20.50 years; b).

Table 2
Nordic strength data (N) for the dominant and non-dominant limbs of the No Injury group and for the injured and contralateral uninjured limbs for the Hamstring Injury group.

	No Injury (n = 43)		Hamstring Injury (n = 11)[a]	
	Dominant	Non-Dominant	Injured	Uninjured
Mean (SD)	405 (105)	399 (111)	402 (71)	422 (89)
Range (Min to Max)	176 to 580	146 to 566	235 to 513	222 to 581

[a] Excludes players with a history of bilateral hamstring injury, Min = Minimum, Max = Maximum.

players who reported never sustaining a hamstring injury. In the present study, a history of hamstring injury did not significantly influence Nordic strength in AFL players, independent of age as a covariate. Previous studies investigating either entirely or partly AFL athletes have demonstrated either no residual deficit, or a 15% residual deficit in Nordic strength after hamstring injury (Opar et al., 2013, 2015a), respectively. Deficits in Nordic strength reported in other athletic cohorts following a hamstring injury have ranged from no deficit (Giakoumis et al., 2020), up to approximately 7% (Vicens-Bordas et al., 2020), and 9% (35 N) (Ribeiro-Alvares et al., 2021). These differences may be due to the calibre of athletes being

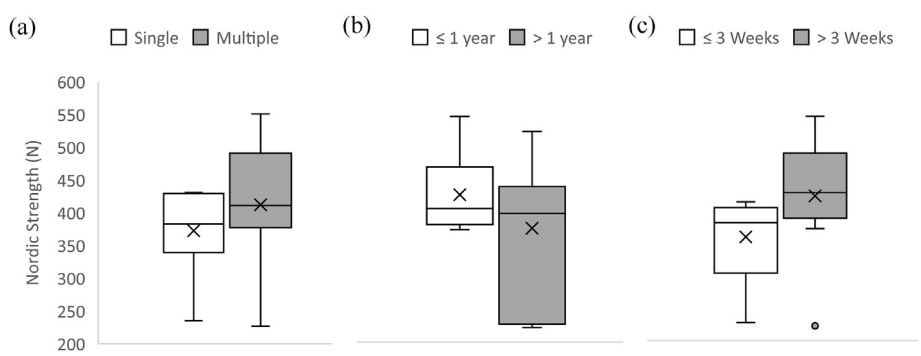

Fig. 2. Boxplots of Nordic strength data for the injured limb of each individual sub-group. No significant differences were demonstrated between any sub-group comparison (i.e., single and multiple episodes (a), ≤ 1-year and > 1-year since last injury (b), and ≤3 weeks and >3 weeks injury severity (c)).

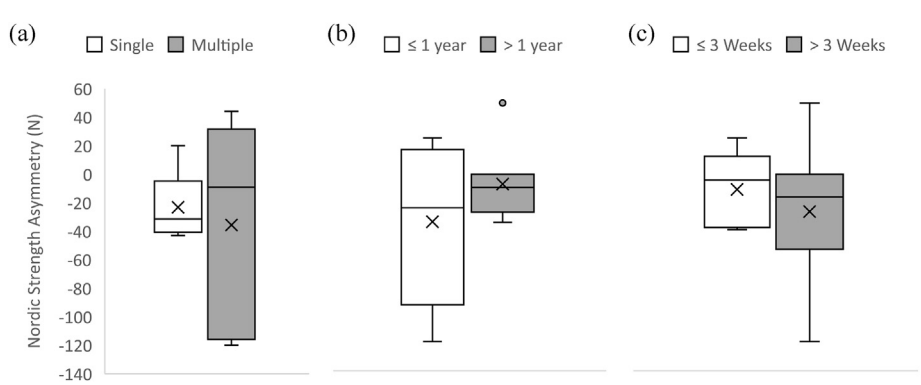

Fig. 3. Boxplots of between limb asymmetry (injured limb minus contralateral uninjured limb) in Nordic strength (N) for each individual sub-group of Hamstring Injury, i.e., number of episodes (a), time since last injury ≤ 1-year and > 1-year ago (b), and injury severity ≤3 weeks and >3 weeks (c). No significant differences were demonstrated between any sub-group.

Table 3
Nordic strength data (N) for the injured and contralateral uninjured limbs for the Hamstring Injury sub-groups.

Sub-group	Limb	Median (IQR)	Range (Min to Max)
Single (n = 6)	Injured	383 (90)	235 to 431
	Uninjured	403 (116)	222 to 456
Multiple (n = 5)	Injured	433 (104)	367 to 513
	Uninjured	469 (119)	414 to 581
≤1 year (n = 8)	Injured	399 (86)	367 to 535
	Uninjured	425 (121)	356 to 581
>1 year (n = 6)	Injured	400 (226)	222 to 513
	Uninjured	422 (225)	232 to 469
≤3 weeks (n = 5)	Injured	376 (98)	227 to 407
	Uninjured	414 (131)	232 to 433
>3 weeks (n = 10)	Injured	421 (97)	222 to 535
	Uninjured	449 (92)	235 to 581

investigated, or higher strength values achieved. Studies in which no residual weakness in Nordic strength after a hamstring injury was found included participants who were world-class athletes within their chosen sport (Giakoumis et al., 2020; Opar, Williams, et al., 2015; Roe, Malone, et al., 2018), compared with results from amateur (Vicens-Bordas et al., 2020) or non-world class athletes (Ribeiro-Alvares et al., 2021) who did demonstrate residual weakness. It is possible that higher calibre athletes may have undertaken a more rigorous, focussed, and lengthier rehabilitation, however, details of rehabilitation were not provided. Alternatively, it may be suggested that achieving higher Nordic strength values (as seen within our study, mean values for all groups 383–411 N) may have minimized any strength deficits post hamstring injury. These Nordic strength values, along with the finding of a lack of residual strength deficits post hamstring injury are similar to those reported by Giakoumis et al. (2020) of 409 N for elite male track and field athletes. Additionally, some studies that demonstrated residual deficits post hamstring injury also had lower Nordic strength values, such as 289 N in predominantly AFL players (Timmins et al., 2015), and 351 N in professional soccer players (Ribeiro-Alvares et al., 2021). The greater mean Nordic strength values observed in our study may reflect adaptations to strength training, including both specific neurological adaptations and muscle hypertrophy (Narici et al., 1996; Seymore et al., 2017), and the improved Nordic strength of AFL players since the original study of Opar, Williams, et al. (2015) are most likely due to changes in training and rehabilitation practices within the AFL.

4.2. Between limb differences in Nordic strength

We found no significant differences in Nordic strength asymmetry between the injured and uninjured contralateral limbs for any group with a history of hamstring injury, and no differences between the dominant and non-dominant limbs for the uninjured group. Our findings of limb asymmetry for the previously uninjured group of 1.5% (or 7 N), are less than those found in a recent meta-analysis of limb asymmetry from previously uninjured athletic individuals of <6% (isokinetic hamstring strength) (McGrath et al., 2016), and <10% from previously uninjured elite Gaelic footballers (Nordic strength) (Roe, Malone, et al., 2018). In the present study, the mean absolute strength of the previously injured limb was always lower compared with the uninjured contralateral limb, but these differences were not statistically significant. This asymmetry, however, ranged from 4 to 9%, which was similar to values reported previously in uninjured athletic individuals (McGrath et al., 2016), and previously uninjured elite Gaelic footballers (Roe, Malone, et al., 2018). The asymmetry observed in the current study is therefore within normal ranges and is unlikely to be clinically relevant or related to the previous injury. When relative strength data were analyzed, the previously injured limb was found to be weaker compared with the uninjured contralateral limb, except for the >3 matches missed sub-groups, however these differences were not statistically significant. The small level of asymmetry found within this study is also unlikely to be relevant to hamstring re-injury risk, however further research is needed to confirm or refute this.

4.3. Effect of number of episodes, time since injury and injury severity, on Nordic strength

In contrast to our hypothesis, the number of injury episodes, time since the last injury, and injury severity, had no significant influence on Nordic strength or strength asymmetry. To our knowledge, this is the first study that collectively investigated Nordic strength in these different subgroups, which makes comparison to the wider literature difficult. Of the studies that subgrouped hamstring injury based on severity, results showed that the limb that sustained either a mild or moderate injury did not have decreased Nordic strength compared with players with no previous injury (Roe, Malone, et al., 2018; Vicens-Bordas et al., 2020). However, for more severe injuries, no significant between limb weakness was found in elite Gaelic footballers with a history of hamstring injury of >28-days (Roe, Malone, et al., 2018), whereas a 9% (age adjusted) reduction in Nordic strength within the previously injured limb compared with players without a prior hamstring injury in amateur soccer players with a > 3-week injury severity has been demonstrated (Vicens-Bordas et al., 2020).

Considering the moderate-to-high effect sizes when comparing the previously injured limb of each subgroup within the present study, and that the between group strength differences for some of the subgroup analysis (single versus multiple hamstring injuries, 50 N; and ≤3 versus >3 matches missed, 45 N) were still within the minimal detectable change of 61–68 N as outlined by Opar et al. (2013), it is unclear if actual differences exist between some subgroups. However, it should be acknowledged that the small numbers of injuries in the subgroups may have inflated the risk of a type II error. Further research with greater subgroup numbers is required to ascertain if the strength differences between these subgroups are confirmed or refuted. Given that the participants within this study were professional athletes and are more likely to have a structured, rigorous outcome-based rehabilitation program, these findings may be explained by improved training and rehabilitation of these injuries in the AFL in recent years. Additionally, all AFL teams within this study consistently performed the NHE, as well as other interventions, as part of their hamstring injury prevention and rehabilitation programs, and this may have contributed to the lack of significant between group findings. The results of the current investigation may not be transferable to lesser trained individuals. It appears that, given appropriate rehabilitation, the number of previous hamstring injuries, the time since last injury, and the severity of the injury do not significantly influence Nordic strength values, or asymmetry, and they can be restored to those of previously uninjured players.

4.4. Limitations

The main limitation of this study was the low numbers of players within the subgrouping analysis with a previous hamstring injury, which may have underpowered this study. However, with the observed effect size of 0.26 for percentage asymmetry, an

adequately powered study would require more than 300 previously injured participants to be matched to more than 300 healthy athletes to achieve statistical significance. However, with such large numbers, it is unlikely a study of this magnitude will ever be done in a homogenous cohort. The self-reported questionnaire, which was used to determine history of hamstring injury, relies on memory recall, and the details of injury specifics from many years' prior is a limitation. Furthermore, the type and amount of training and rehabilitation performed by participants was not controlled and therefore may have differed between clubs and individual players. It is important to note, that these results are based on elite athletes and may not be transferable to non-professional athletes, who most likely undertake less rigorous rehabilitation. Finally, although Nordic strength was assessed on the same type of device (Nordbord, Vald Performance), testing was conducted across four different sites, by different assessors and on different devices. However, the reliability of this technique in the assessment of bilateral maximal strength has been shown to be high i.e. ≥ 0.85 (Opar et al., 2013) which gives us confidence that our findings are robust.

5. Conclusions

AFL players with a history of hamstring injury did not exhibit deficits in Nordic strength relative to their uninjured contralateral limbs or players without a previous hamstring injury. Nordic strength was also not significantly influenced by the number, recency, or severity of hamstring injury in elite AFL players, however, some potential clinical differences may be apparent and warrant further investigation. Future efforts aimed at preventing hamstring injury and its recurrence should investigate multifactorial injury mechanisms and ensure adequate subject numbers for these interactions.

Ethical approval

This work was approved by the relevant ethical committee related to the University in which the research was performed. All subjects provided their written informed consent prior to their participation in the study.

Declaration of competing interest

Dr Matt Bourne is undertaking an Australian Federal Government funded fellowship in collaboration with Vald and has previously received funding from Vald for research unrelated to the current study. Dr Matt Bourne and Professor Rod Barrett are recipients of a research grant from a commercial entity that was paid to the University for a project unrelated to the present study. Vald Performance are global distributors of the Nordbord. The remaining authors declare no conflicts of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ptsp.2022.06.006.

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